

Continuous Casting Machine (CCM) - Standard Operating Procedures

W and S Unit Hydraulic Valve Replacement

- Step 1.** Verify the continuous caster strand is fully stopped and the dummy bar is mechanically secured.
- Step 2.** Execute Lockout Tagout on the main Hydraulic Power Unit powering the W and S zone.
- Step 3.** Isolate the local nitrogen bladder accumulator for the specific pinch roll segment.
- Step 4.** Manually bleed residual line pressure via the manifold dump valve until the local analog gauge reads zero bar.
- Step 5.** Disconnect the electrical M12 connectors from the valve's solenoids and the LVDT position sensor.
- Step 6.** Unbolt the proportional directional valve from the manifold block using a cross pattern sequence to prevent thread binding.
- Step 7.** Inspect the manifold mating surface for scoring, and verify all old O rings are removed.
- Step 8.** Mount the replacement valve with new O rings, torquing the bolts to the specified limit in a cross pattern.
- Step 9.** Reconnect the solenoid and LVDT electrical connectors.
- Step 10.** Remove Lockout Tagout, slowly open the isolation block to pressurize the segment, and actuate the manual override to bleed trapped air.
- Step 11.** Confirm roll gap calibration and valve stroke response via the SCADA interface.